TREADING WATER

INDIA'S SUBMARINE FLEET



- The first submarine, INS *Kalvari*, Soviet Foxtrot-class was commissioned in Dec. 1967
- The Indian Navy had 21 such conventional submarines in the early 1980s
- Seven Russian Kilo Class or Sindhughosh Class, bought from Russia between 1984 and 2000
- Four HDW (German) subs, known as Shishumar Class, bought and then built in India in collaboration with the Germans (1986-1994)
- Four Kalvari Class Scorpene submarines (under Project-75) built at Mazagon Dock in partnership with France's Naval Group commissioned. Two more to be delivered soon
- An Arihant SSBN, a slow-moving 'bomber' and a stealth launch platform for N-weapons commissioned. Three more SSBNs under construction, part of the Strategic Forces Command

PAKISTAN

- Pak navy operates three Agosta-90B submarines from France's Naval Group
- Signed a contract with China in 2015 for eight Yuan-class (Type 039-A) conventional air-independent propulsion-capable submarines (cost: $5 billion)



CHINESE PLA-N

- PLA Navy operates 59 diesel-powered subs of Yuan-Class (Type 039A or Type 041),Kilo-Class, Song-Class (Type 039) & Ming Class (Type 035)
- Six Shang-class (Type 093B) N-powered attack submarines
- Four Jin-class (Type 094/094A) nuclear-powered ballistic missile submarines



ANI

Russian-origin kilo-class submarine, INS *Sindhudhvaj*, was decommissioned after 35 years of service. Incidentally, all the Indian submarines, except for the four new Scorpene class ones, are three decades old. While four of the available Russian kilo class submarines are undergoing upgrades for extended service life, the remaining three will have run their course by the end of the decade.

The Indian submarine fleet is tasked with defending a 7,600 km coastline and patrolling a sphere of influence that stretches from the Gulf of Aden to the Malacca Straits for intelligence-gathering, anti-submarine warfare, anti-surface warfare and mine-laying operations.

Naval experts believe that with the mid-life upgrades, the navy is desperately trying to maintain minimal underwater capability. But these submarines, even after upgrades, will be facing an enemy submarine or anti-submarine capability of the present generation. The majority of Indian operational submarines were designed in the 1970s-80s.

Submarines play a key role in all three aspects of maritime warfare i.e. sea control, sea denial and maritime power projection. They provide 'sea control' with their firepower and sensors, while the 'sea denial' element focuses on preventing enemy warships from entering Indian waters.

Ideally, the navy should have had at least 24 submarines by now to meet its 30-year submarine building plan (approved by the cabinet committee on security in 1999 after the Kargil conflict). The plan was to induct 12 diesel subs by 2012 and another 12 submarines by 2030, but repeated delays forced the Navy to rejig the plan. Now, the plan projection is 18 diesel-powered submarines and six SSNs (N-power subs) by 2030. The purpose of having a 30-year plan was to develop a submarine-building capability that would keep delivering to the Indian navy. With the replacement of old decommissioned submarines, the Indian navy, at any given time, should have (or operate) at least a 20-strong submarine force. So, Project 75 and Project 75 (India)—to build six conventional submarines each—were designed to run concurrently. Two established submarine production lines would ensure that one sub could get commissioned every alternate year. But it never happened.

In 2006, an agreement between the French firm Naval Group (formerly DCNS) and Mazagon Dock Ltd (MDL) was signed to build six Scorpene class submarines in India. The first submarine was scheduled for delivery in 2012, but the project witnessed repeated delays. Four Scorpene submarines have been delivered so far, with two more to go.

The next programme, Project 75 (India), with an estimated cost of Rs 43,000 crore was to build six conventional submarines with better sensors and weapons and AIP in the country. Air Independent Propulsion or AIP is a marine propulsion technology that allows non-nuclear submarines to operate without access to atmospheric oxygen. It allows submarines to stay underwater for longer periods. AIP-fitted submarines also do not have to surface to charge their batteries (they are classified as SSP while other diesel attack submarines are known as SSKs). Though the programme was initiated way back in November 2007,